



# Automatic generation of Slovenian traffic news for RTV Slovenija

Miha Lazić, Luka Gulič, and Matevž Crček

## Abstract

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## Keywords

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Advisors: Slavko Žitnik

## Introduction

Traffic reporting is a crucial component of public broadcasting, providing real-time updates on road conditions, accidents, and congestion. At RTV Slovenija, traffic news is currently compiled manually by students who review traffic reports from the *promet.si* portal and transcribe them into structured news segments, which are then broadcast every 30 minutes. This manual approach is time-consuming, prone to human error, and inefficient given the volume of traffic data that needs to be processed.

With the advancement of large language models (LLMs), there is an opportunity to automate the generation of structured and readable traffic news. This project aims to leverage an existing LLM, apply prompt engineering techniques and fine-tune the model to automatically generate traffic reports that adhere to the broadcasting standards of RTV Slovenija. In the next chapter, we present various existing solutions that we consider using in this project.

## Related Work and Existing Solutions

In their master's thesis, Bjørge and Lundegaard explored the automation of traffic incident reporting. They developed a system that collects real-time traffic data, processes it to identify incidents, and automatically generates reports for dissemination to the public and relevant authorities. Their work provides a detailed description of approaches that could be highly beneficial to our project [1].

## LLM Selection

For this project, we consider three potential LLMs: *LLaMA*, *FLAN-T5*, and *Bloom*. These models have been pre-trained

on multilingual datasets, including Slovenian text, enabling them to process Slovenian-language prompts. However, none of these models have been specifically trained on Slovenian traffic data, making fine-tuning essential for achieving optimal performance.

Both *LLaMA* and *FLAN-T5* demonstrate strong reasoning capabilities, making them well-suited for structured traffic report generation. In contrast, *Bloom* is less effective in reasoning tasks, which may lead to lower performance when applying *Chain-of-Thought Prompting*.

For evaluation, we think about combination of automatic and semi-automatic evaluation criteria. Automatic evaluation quantitatively assesses LLM outputs without human intervention. For this, we employ algorithms such as *BLEU*, which measures similarity to reference texts, and *ROUGE*, which evaluates recall-based overlap. However, for a more comprehensive assessment, human involvement is essential. Therefore, we incorporate semi-automatic evaluation, leveraging AI-assisted scoring. Specifically, we use another LLM, to grade generated outputs based on predefined criteria. Each score is then manually reviewed and adjusted as needed to ensure accuracy.

## Prompt Engineering

To generate traffic reports, we consider using a combination of *Role-Based Prompting* and *Few-Shot Prompting*. This approach is particularly suitable because our dataset consists of structured traffic data paired with human-written traffic reports. By assigning the LLM the role of a traffic reporter and providing it with well-structured example reports, we aim to guide the model towards generating high-quality outputs.

Before report generation, the model will be given explicit

instructions on how to structure the reports, which include:

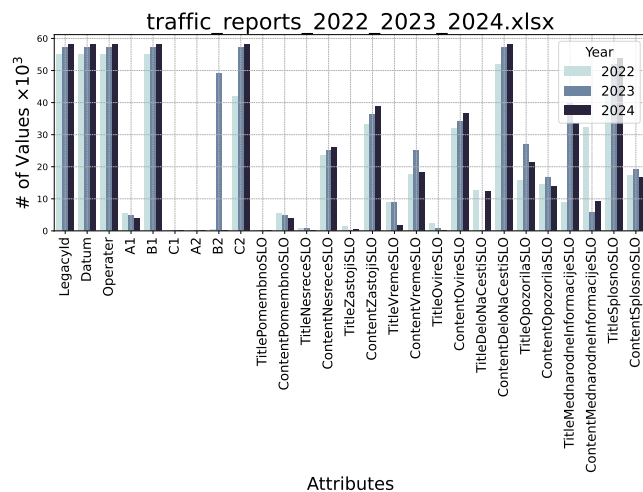
- The location of the incident, specifying the affected road and direction.
- The cause of the disruption and its consequences.
- The specific section of the road impacted.

Additionally, the dataset contains road names that differ from those used in official reports, requiring the model to handle these discrepancies appropriately.

To ensure proper prioritization of events, an event hierarchy must be established, as some incidents are more critical than others. If the model fails to respect this hierarchy, we will experiment with *Chain-of-Thought Prompting* before fine-tuning the model. This technique introduces step-by-step reasoning, helping the model determine which incidents should be reported first.

### Data Analysis

The primary dataset is the `traffic_reports_2022_2023_2024.xlsx` file from the *promet.si* portal, which contains 170,823 traffic reports from the years 2022, 2023, and 2024, organized into 27 attributes, as illustrated in Figure 1, following with their descriptions.



**Figure 1.** Project's primary dataset visualization.

**LegacyId** Unique numerical identifier for the traffic report.

**Datum** Timestamp of when the traffic report was logged.

**Operator** Name and affiliation of the traffic report operator.

**A1** Very important traffic alert message in Slovenian.

**B1** Important traffic information in Slovenian.

**C1** Less important information in Slovenian.

**A2, B2, C2** English versions of A1, B1, C1.

**Title+ContentPomembnoSLO** Title and content of important Slovenian traffic information.

**Title+ContentNesreceSLO** Title and content related to road accidents.

**Title+ContentZastojiSLO** Title and content describing traffic congestion or delays.

**Title+ContentVremeSLO** Title and content related to weather conditions or weather-related traffic impacts.

**Title+ContentOvireSLO** Title and content concerning traffic obstacles (vehicle breakdowns, road hazards).

**Title+ContentDeloNaCestiSLO** Title and content related to roadworks or construction affecting traffic.

**Title+ContentOpozorilaSLO** Title and content containing public warnings, announcements, or safety reminders.

**Title+ContentMednarodneInformacijeSLO** Title and content regarding international or cross-border traffic information.

**Title+ContentSplosnoSLO** Title and content of general traffic information not covered by other specific categories.

## Methods

Use the Methods section to describe what you did and how you did it – in what way did you prepare the data, what algorithms did you use, how did you test various solutions ... Provide all the required details for a reproduction of your work.

Below are  $\text{\LaTeX}$  examples of some common elements that you will probably need when writing your report (e.g. figures, equations, lists, code examples ...).

### Equations

You can write equations inline, e.g.  $\cos \pi = -1$ ,  $E = m \cdot c^2$  and  $\alpha$ , or you can include them as separate objects. The Bayes's rule is stated mathematically as:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}, \quad (1)$$

where  $A$  and  $B$  are some events. You can also reference it – the equation 1 describes the Bayes's rule.

### Lists

We can insert numbered and bullet lists:

1. First item in the list.
  2. Second item in the list.
  3. Third item in the list.
- First item in the list.
  - Second item in the list.
  - Third item in the list.

We can use the description environment to define or describe key terms and phrases.

**Word** What is a word?.

**Concept** What is a concept?

**Idea** What is an idea?

### Random text

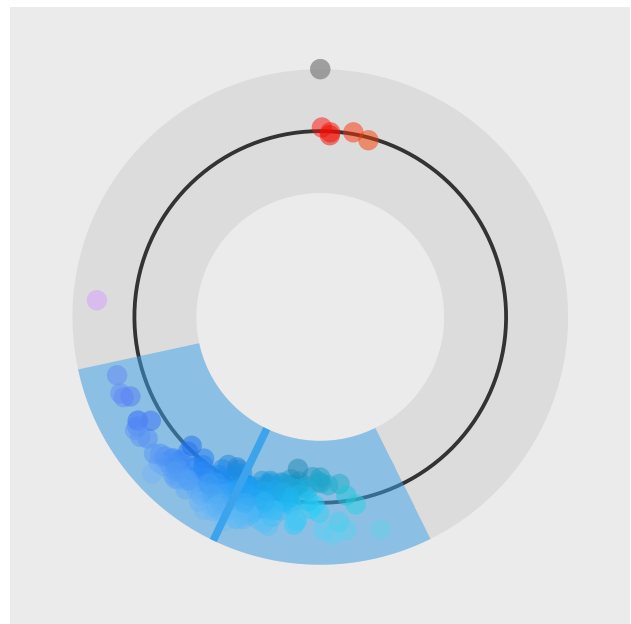
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### Figures

You can insert figures that span over the whole page, or over just a single column. The first one, Figure 2, is an example of a figure that spans only across one of the two columns in the report.



**Figure 2. A random visualization.** This is an example of a figure that spans only across one of the two columns.

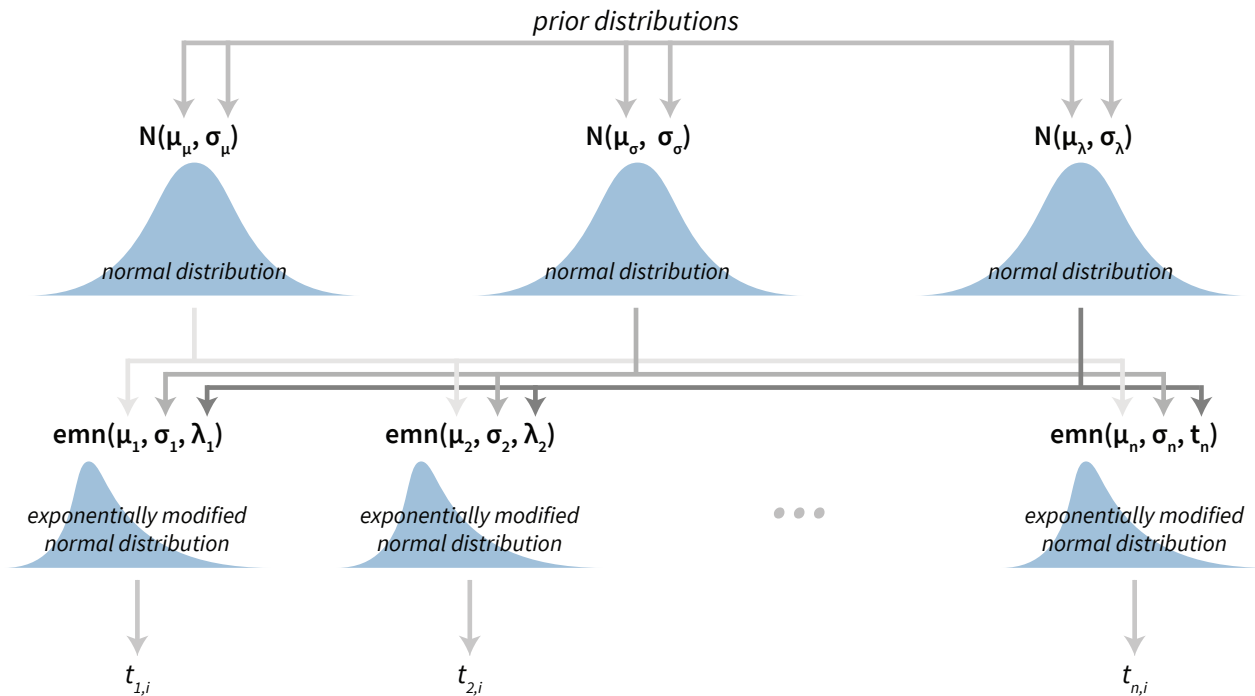
On the other hand, Figure 3 is an example of a figure that spans across the whole page (across both columns) of the report.

### Tables

Use the table environment to insert tables.

**Table 1.** Table of grades.

| Name       |           | Grade |
|------------|-----------|-------|
| First name | Last Name |       |
| John       | Doe       | 7.5   |
| Jane       | Doe       | 10    |
| Mike       | Smith     | 8     |



**Figure 3. Visualization of a Bayesian hierarchical model.** This is an example of a figure that spans the whole width of the report.

### Code examples

You can also insert short code examples. You can specify them manually, or insert a whole file with code. Please avoid inserting long code snippets, advisors will have access to your repositories and can take a look at your code there. If necessary, you can use this technique to insert code (or pseudo code) of short algorithms that are crucial for the understanding of the manuscript.

#### Listing 1. Insert code directly from a file.

```
import os
import time
import random

fruits = ["apple", "banana", "cherry"]
for x in fruits:
    print(x)
```

#### Listing 2. Write the code you want to insert.

```
import (dplyr)
import (ggplot)

ggplot (diamonds,
        aes(x=carat, y=price, color=cut)) +
  geom_point() +
  geom_smooth()
```

## Results

Use the results section to present the final results of your work. Present the results in a objective and scientific fashion. Use

visualisations to convey your results in a clear and efficient manner. When comparing results between various techniques use appropriate statistical methodology.

### More random text

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## Discussion

Use the Discussion section to objectively evaluate your work, do not just put praise on everything you did, be critical and

exposes flaws and weaknesses of your solution. You can also explain what you would do differently if you would be able to start again and what upgrades could be done on the project in the future.

## Acknowledgments

Here you can thank other persons (advisors, colleagues ...) that contributed to the successful completion of your project.

## References

- [1] Anna Jansdatter Bjørge and Mads Lundegaard. Automating the process of traffic incident reporting. Master's thesis, NTNU, 2024.